Eliminating Spatial Correlations of Anomaly: Corner-Visible Network for Unsupervised Hyperspectral Anomaly Detection

Leiquan Wang, *Member, IEEE*, Yunfeng Wang, Zhongwei Li, Chunlei Wu, *Member, IEEE*, Mingming Xu, *Member, IEEE*, and Mingwen Shao

***Abstract*— Hyperspectral anomaly detection (HAD) is crucial for identifying and analyzing abnormal objects in various domains. While existing methods have shown promising results by designing detection methods tailored to specific anomaly characteristics, there is a need for a highly versatile approach that can effectively handle anomalies, particularly those with large spatial sizes. In this article, we propose an end-to-end corner-visible network (CVNet) for unsupervised HAD. Specifically, we introduce a corner-visible convolution that leverages the statistical dependencies of the background within the receptive field while eliminating spatial correlations with potential anomalies for background generation. To address the grid effect caused by the corner-visible convolution, a background smoothing module is employed by using the conventional convolution. Furthermore, adaptive mean-squared error (MSE) and structural similarity index (SSIM) losses are employed to suppress anomaly reconstruction, resulting in a reliable reconstructed background map. Anomalies are identified through the residual of the original hyperspectral image (HSI) and the reconstructed background. Extensive experiments conducted on three publicly datasets demonstrate the effectiveness of our proposed method in handling different types of anomalies. The state-of-the-art performance showcases the versatility and applicability of CVNet in HAD. The codes of this work will be available for the sake of reproducibility at https://github.com/Cloudynewbee/CVNet-HAD.**

***Index Terms*— Background reconstruction, corner-visible convolution, hyperspectral anomaly detection (HAD), spatial correlation.**

I. Introduction

REMOTE sensing images offer advantages including wide coverage, high spatial resolution, timely observations, multispectral information, and high efficiency, all of which contribute to their broad range of application value [1], [2]. In particular, hyperspectral images (HSIs) capture the spectral signature by recording hundreds of narrow wavelength bands within each spatial pixel [3]. The high-resolution spectral information provides a tool for precise Earth observation [4], [5]. Unlike anomaly detection in natural images, which primarily emphasizes visually distinctive anomalies, hyperspectral anomaly detection (HAD) necessitates the consideration of spectral deviations in conjunction with visual characteristics. This holistic approach enables the identification of subtle anomalies that could be overlooked by alternative imaging techniques, rendering HAD a valuable asset across a diverse array of fields, including agricultural yield estimation [6], mineral detection [7], environmental protection [8], cultural relic restoration, military camouflage, and recognition [9].

In recent years, a variety of HAD methods have been developed. Traditionally, HAD has been approached from a statistical perspective, where global or local statistical models are used to estimate the background distribution. For example, the Reed–Xiaoli (RX) algorithm [10] assumes that the background follows a Gaussian distribution and identifies abnormal targets based on their deviation from the background. However, it is important to note that HSIs may not always conform to the assumptions of these statistical models, which can limit their effectiveness in certain scenarios.

Representation-based methods in HAD attempt to represent the background as a linear combination of dictionary elements, typically by constructing a dictionary model. For example, the collaborative representation-based detector (CRD) algorithm [11] employs a sliding dual window sampling technique to build a dictionary. It then calculates adaptive weights using a regularized collaborative presentation of a similarity matrix and estimates background pixels via a linear combination of samples. Abnormal levels are identified through the residuals. However, representation-based methods may suffer from high computation costs during matrix decomposition, which can further limit their effectiveness. Additionally, a drawback of representation-based methods is that they often do not take into account the local correlation of spatial information.

HSIs not only provide high-density spectral information but also preserve the spatial information of ground objects [12]. Distinct land cover types can possess similar spectral attributes, such as the green appearance of vegetation types like grass and trees, while the same land cover type can exhibit

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Leiquan Wang, Yunfeng Wang, Chunlei Wu, and Mingwen Shao are with the Qingdao Institute of Software, College of Computer Science and Technology, China University of Petroleum, Qingdao 266580, China (e-mail: richiewlq@gmail.com; wwang1491@163.com; wuchunlei@upc.edu.cn; smw278@126.com).

Zhongwei Li and Mingming Xu are with the College of Oceanography and Space Informatics, China University of Petroleum, Qingdao 266580, China (e-mail: lizhongwei@upc.edu.cn; xumingming@upc.edu.cn).

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different spectral features influenced by factors such as varying illumination and weather conditions [8]. The use of spectral information alone for anomaly detection is subject to limitations. Moreover, specific land features, such as buildings and roads, exhibit a consistent spatial arrangement that sets them apart from other land features. Spatial–spectral methods use a series of spatial attribute filtering to separate the background and anomaly. The anomaly extraction and detection method [13], for instance, uses morphological attribute filtering to extract anomaly candidates and then refines anomaly features through edge-preserving filtering to obtain the final anomaly image. However, spatial filtering operations are often used during image pre-processing or post-processing, but they do not typically consider the deep features contained in the data.

In recent years, deep learning has shown significant promise in the field of hyperspectral imaging analysis, including in the realm of HAD. Convolutional neural networks (CNNs) have emerged as effective tools for extracting deep features and have been extensively applied in various domains of remote sensing image analysis [14], [15], [16]. The method called convolutional neural network detector (CNND) [17] introduced deep CNNs to HAD for the first time. This method uses labeled data to create pixel pairs for training the deep CNN. During testing, the trained model takes pixel pairs composed of the test pixel and its surrounding pixels as input to calculate similarity, and the average similarity represents the abnormal level. However, due to its supervised training mode, this method requires labeled data, which may only be partially available or not available at all in many real-world systems, limiting the applicability of this approach.

To overcome the limitations of supervised methods, researchers have shifted their focus toward unsupervised deep HAD methods, which assume that anomalies are rare and significantly different from normal instances. Autoencoders (AEs), generative adversarial networks (GANs), and their variants have been widely used as major network structures for unsupervised HAD. They learn the spectral characteristics of the HSI background and reconstruct the spectral vector pixel by pixel. Residual processing is then carried out between the reconstruction results and the original image, where the reconstruction error is considered the abnormal level. BASGAN [18] obtains the pseudo-background vector through rough extraction during training and then uses the trained model to reconstruct the original HSI. Since no abnormal vector is introduced during training, the reconstructed result is only representative of the background. In contrast, the authors of SLDR [19] leverage AE to detect anomalies for HSI. A constraint is imposed on the encoder to encourage the generation of latent variables that follow a Gaussian distribution. This constraint ensures that the spectral information of the original image is preserved and helps to improve the accuracy of anomaly detection. Moreover, self-supervised anomaly detection techniques have garnered significant attention in recent years [20]. Notably, PDBSNet [21] and BS3LNet [22] have effectively integrated self-supervised learning strategies into the domain of HAD, yielding promising outcomes. Nevertheless, such methods remain contingent on the availability of supervised signals, thereby constraining their applicability to particular contexts.

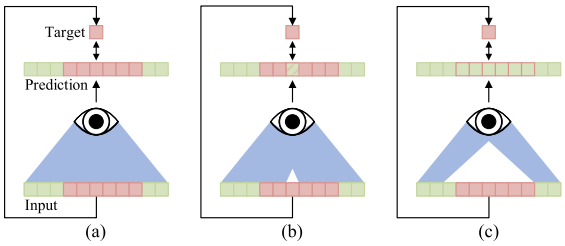


Fig. 1. Comparisons between conventional convolution network, blind-spot network, and CVNet. (a) In the conventional convolution network, all elements within the receptive field are utilized for prediction. (b) In the blind-spot network, the receptive field of each pixel excludes the pixel itself, thereby preventing it from learning its own identity. (c) In the CVNet, only corner pixels located far away from the center of the receptive field are employed, eliminating the spatial correlation within the central region.

The above methods work on the pixel-wise spectral domain, where the spatial information is almost ignored in the embedding process. To address this limitation, Auto-AD [23] conducts 2-D convolution to reconstruct the background with an adaptive loss function to suppress abnormal reconstruction. However, the excessive downsampling operations performed during the encoder phase lead to a reduction in spatial resolution, resulting in the loss of valuable details required for background reconstruction. In addition, the conventional convolution operator tends to learn the identity relationship of candidate pixels [24], placing undue emphasis on the feature learning between the background and anomalies. Drawing from the experience of image denoising, the authors of BS^{3}LNet [22] have introduced the blind-spot neural network to HAD, with a focus on achieving faithful background reconstruction. However, while the adopted blind-spot neural network, originally proposed for image denoising, proves effective for small-scale anomalies, it faces limitations when dealing with a wider range of anomalies. The crux of the matter lies in the fact that the blind-spot neural network maximizes the dependencies among background pixels but overlooks the correlations of anomalies. As a result, it may inadvertently capture excessive abnormal information within the local receptive field, which negatively impacts the accuracy and fidelity of the background reconstruction. The spatial correlation is a key issue for HAD, however, has not been well investigated in HAD so far. The article aims to address this challenge by exploring ways to eliminate the spatial correlations of anomalies, adapting the image denoising approach to better suit the requirements of HAD. Fig. 1 illustrates the differences between the conventional convolution, blind-spot convolution, and our corner-visible convolution.

The key issue of HAD, therefore, is how to reconstruct a central background pixel using a specific convolution operation that exploits the spatial correlations of the background while eliminating the spatial correlations of the anomalies. In this article, an end-to-end background reconstruction method with corner-visible network (CVNet) is proposed for HAD by learning the background spatial correlations, where the anomalies appear as prominent reconstruction errors. The spatial

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correlations of the background are learned by mapping a uniform noise space onto the hyperspectral space. To preserve the spatial information, the size of the feature map is kept consistent with that of the original HSI during the entire feature extraction process. A corner-visible architecture is proposed for background reconstruction, which enhances the statistical dependencies of the background while eliminating spatial correlations with anomalies. The CVNet utilizes a large masked receptive field to prevent the network from learning the identity of anomalies, resulting in the anomalies not being reconstructed. An adaptive loss function is performed to further suppress anomaly reconstruction.

The main contributions of this article can be summarized as follows.

1) An end-to-end background reconstruction method with CVNet is proposed for unsupervised HAD. Anomalies are detected by adapting the image denoising principles to meet the specific requirements of HAD, enabled by spatial correlation hypothesis on background and anomaly pixels.
2) A corner-visible convolution is proposed by utilizing the statistical dependencies of the background within the receptive field while eliminating spatial correlations with anomalies. The CVNet enhances the background reconstruction while keeping the anomalies not being reconstructed.
3) The adaptive-weighted structural similarity index (SSIM) loss is introduced for the first time in the context of HAD tasks to preserve fine-grained spatial details in the reconstructed backgrounds. By combining the adaptive SSIM and mean-squared error (MSE) losses, the weights of potential anomalous pixels are suppressed, resulting in precise and reliable background reconstructions.
4) Extensive experiments were conducted on the three available HSI datasets to evaluate the performance of our method. The experimental results demonstrate that our method exhibits exceptional versatility, allowing it to be applied to different types of anomalies. Furthermore, our method achieves state-of-the-art accuracy for unsupervised HAD.

Section II provides an overview of related work in the field. Section III presents the proposed method, detailing its framework and key components. In Section IV, we present the experimental results and analyze the performance of our approach, comparing it to other existing methods. Finally, Section V concludes the article, summarizing the main findings and discussing future research directions.

II. Related Work

Currently, traditional methods for HAD have shown satisfactory performance, including statistical-based methods [25], [26], [27], representation-based methods [28], [29], [30], and pixel-wise detection methods [31], [32]. However, due to the spectral redundancy and noise in HSIs, these methods do not perform optimally in detecting anomalies in certain scenes. On the other hand, deep learning-based methods have demonstrated promising results in recent years by leveraging the deep features of hyperspectral data.

A. CNN and Transformer-Based Methods

CNND [17] was the pioneer in utilizing deep CNNs for HAD. Despite the reliance on strong sample support during training, it showcased the advantageous handling of spatial features by CNNs in this context. For the first time, Xiao et al. [33] applied transformer to HAD. They initially utilized convolutional aggregation of local pixels to fuse similar spectral vectors. Subsequently, they employed the transformer's ability to extract global information to incorporate a significant amount of background information in order to attenuate abnormal features. Additionally, they achieved favorable results by suppressing anomalies using local dual-window sampling.

B. AE-Based Methods

AE is an unsupervised learning algorithm composed of an encoder and a decoder, which aims to learn a compact and informative representation that captures the most important features of the input data. It is trained by minimizing the reconstruction error, ensuring that the output of the decoder closely approximates the original input data. The reconstruction error of AE serves as an indicator of anomaly level, enabling its successful application in HAD, given the lack of prior knowledge and limited anomaly samples [23], [34], [35], [36], [37], [38]. However, current AE-based methods commonly employ spectral vectors for training, ignoring the spatial features of HSIs.

C. GAN-Based Methods

GANs, built on the foundation of AE, incorporate a discriminator to engage in an adversarial process for optimizing network parameters. This enables GANs to possess the ability to generate vectors similar to input vectors with high precision. However, GANs' training necessitates substantial support from positive samples. In the context of HAD tasks, the abundance of background vectors in HSIs precisely meets this requirement [39], [40], [41], [42]. However, to prevent overfitting, GAN methods require clean background samples for training.

All the aforementioned methods have been validated to achieve localization detection of anomaly objects. However, due to the utilization of different techniques, each method may have its technical limitations, which can potentially affect its detection performance. In contrast, CVNet takes a visual feature-based approach by studying the spatial distribution characteristics of anomalies and backgrounds in HSIs. This approach successfully separates anomalies from the background, leading to a high level of accuracy in HAD.

III. Methodology

A. Formulation

Given an HSI $\mathbf{X} \in \mathbb{R}^{\mathbf{H}\times\mathbf{W}\times\mathbf{C}}$, it can be drawn from the joint distribution

$$p(X) = p(B, A) = p(B) \cdot p(A) \tag{1}$$

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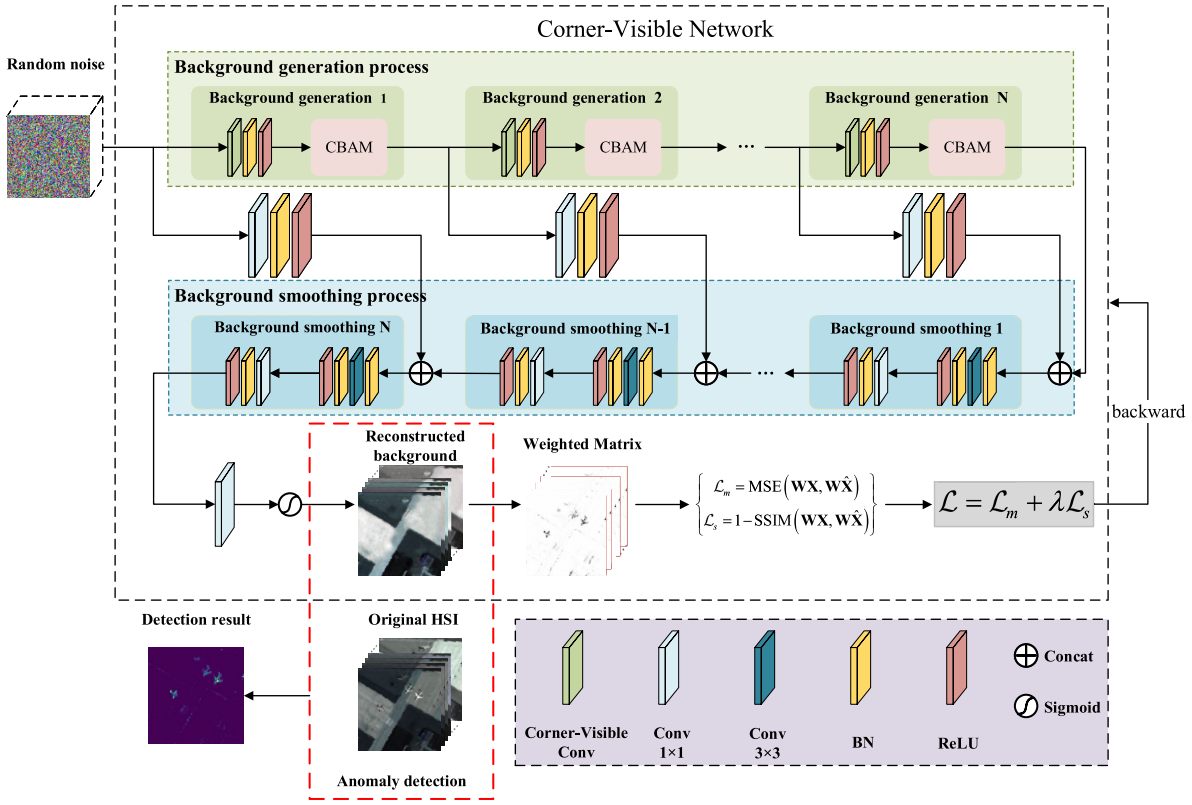


Fig. 2. CVNet for HAD.

where $B = \{b_1, b_2, \ldots, b_m\}$ is the set of background pixels, $A = \{a_1, a_2, \ldots, a_n\}$ is the set of anomalies. Backgrounds B and anomalies A are mutually independent. H and W are the height and width of X. m and n are the numbers of background pixels and anomalies with the constraints $m + n = H \cdot W$ and $m \gg n$. The hypothesis below defines the statistical spatial correlation between pixels.

Hypothesis 1: Background pixels are not pixel-wise independent, exhibiting high spatial correlation among neighboring backgrounds

$$p(b_i|b_j) \neq p(b_i). \tag{2}$$

Anomalies are not pixel-wise independent, exhibiting high spatial correlation among neighboring anomalies

$$p(a_i|a_j) \neq p(a_i). \tag{3}$$

The anomaly pixel is conditionally pixel-wise independent given the background pixel, with negligible spatial correlation between them

$$p(a_i|b_i) = p(a_i) \tag{4}$$

where i and j are two neighborhood pixels within a certain radius.

The hypothesis dictates that the neighbors of the background are not pixel-wise independent, and there is a significant spatial correlation among the background neighbors. The same applies to anomalies. However, the anomalies are pixel-wise independent given the backgrounds. Accounting for the hypothesis, a background b_i can be neatly reconstructed by its background neighbors without the interference of anomalies. Consequently, the HAD task evolved into the background reconstruction, and the final anomaly result can be found through the residual between the original HSI and the reconstructed backgrounds. The key issue is how to make full use of the spatial correlation of background neighbors and try to get rid of the interference of anomalies during the process of background reconstruction.

B. Overview

The primary objective of this article is to reconstruct the background images at the pixel level, ensuring that they have the same size as the original HSI. To accomplish this, a fully convolutional network is employed that maps the input noise space $\mathbf{X}_o \in \mathbb{R}^{H \times W \times C}$ to the hyperspectral background space $\mathbf{X}_{\mathrm{bg}} \in \mathbb{R}^{H \times W \times C}$, where H and W represent the height and width of the image, respectively, and C denotes the number of spectral bands included in the image. By leveraging the correlation among background pixels and eliminating the interference of anomalies, this mapping process effectively captures the spatial–spectral characteristics of the background and generates accurate and realistic reconstructions of the background images. The overall framework of CVNet is illustrated in Fig. 2, with the training process consisting of two distinct stages: background generation and background smoothing.

C. Corner-Visible Convolution

Given a central pixel x_i, we aim to reconstruct it to the background pixel $\hat{x}_i$ by leveraging the spatial correlation of

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its background neighbors. A straightforward approach is to resort to the convolution operations with a certain receptive field $\mathbf{x}_i^f$

$$f\left(\mathbf{x}_i^f;\theta\right)=\hat{x}_i. \tag{5}$$

A pixel's receptive field is defined as a consecutive square patch around that pixel in conventional convolution. The convolution operation, by aggregating pixels within local receptive fields, enables the extraction of high-level semantic features but poses limitations in exploring the low-level relationships between individual pixels. To address the limitation, Gao et al. [22] proposed to use a blind-spot network to detect anomalies in hyperspectral, where the receptive field of each pixel excludes the pixel itself, preventing it from learning the identity. However, the blind-spot network is first proposed for the task of denoising, which is suitable for small-scaled scattered noise. Nevertheless, in HAD, the anomalies are usually unpredictable (adhere to an unknown distribution), where the size of anomalies varies in a diverse manner. From the above consideration, the proposed corner-visible convolution is specifically designed to address HAD. Fig. 3 illustrates the differences between different kinds of convolutions.

As shown in Fig. 3, the receptive field of the convolution operation is positioned at the farthest-visible distance from the central pixel. In this way, we can fully leverage neighboring background pixels and minimize the influence of anomalous pixels. Specifically, the corner-visible convolution is represented as a $k\times k\times c$ convolution that consists of four $k'\times k'\times$c sub-kernels $\mathbf{K}_l, \forall l\in\{1,2,3,4\}$ placed in the corners, where c is the number of input channels, k' is the sub-kernel size, and k' is less than $((k-1)/2)$. During convolution operation, only the input values corresponding to the sub-kernel positions are considered, while the information of the central pixel and its neighborhood is ignored. The convolution result obtained at position (i,j) is represented as

$$\hat{x}^f{}_{i,j}=\sum_{l=1}^{4}\left(\mathbf{K}_l*\mathbf{X}_{i,j,l}\right) \tag{6}$$

where $\mathbf{K}_l$ represents the lth sub-kernel, and $\mathbf{X}_{i,j,l}\in\mathbb{R}^{k'\times k'\times c}$ denotes the lth block within the receptive field centered at coordinates (i,j) that has the same size and position as sub-kernel $\mathbf{K}_l$, $*$ is matrix multiplication.

The corner-visible convolution leverages the correlation of background pixels by incorporating the corner pixels located at the far end of the local receptive field. When encountering anomalous pixels, the corner-visible convolution masks the nearest neighboring pixels, which are likely to be anomalies as well. In contrast, the corner pixels, being less affected by anomalies, are more likely to represent the background. Selective masking helps prevent the correlation of abnormal pixels and ensures that relevant background information is utilized for reconstruction.

D. Background Generation Block

The input noise image generated with an uniform distribution, denoted as $\mathbf{X}_o$, is passed through the consecutive background generation blocks to capture the spatial–spectral correlation of the background. Each block is composed of corner-visible convolution layers, batch normalization (BN) layers, activation layers (ReLU), and convolutional block attention modules (CBAMs) [43].

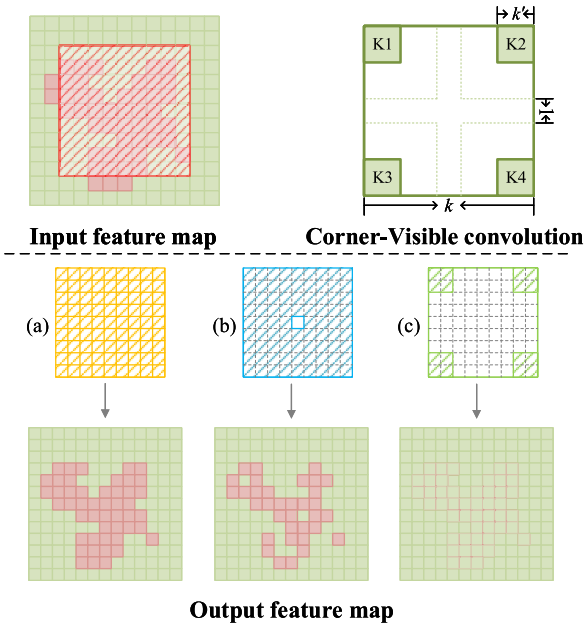


Fig. 3. Illustrations of three different convolution results, where the red grid indicates the anomalies in the input, the shadows represent the receptive field of convolution, and the green dashed box indicates the range of variations in the sub-kernels. (a) Conventional convolution. (b) Blind-spot convolution. (c) Corner-visible convolution.

The corner-visible convolution layer is designed to model the spatial correlations in the background. It employs a large $k\times k$ receptive field to capture the spatial dependencies among neighboring pixels. However, for background reconstruction, only a subset of weights, specifically $4\times k'\times k'$, is utilized, while the remaining weights are masked to prevent the network from learning potential anomalies. The masking mechanism helps ensure that the network focuses solely on reconstructing the background information without being influenced by the presence of anomalies.

After undergoing BN and ReLU activation, the generated features are passed through the CBAM. CBAM is strategically employed to harness both spectral and spatial characteristics, enabling the network to dynamically focus on pertinent features within the input, enhancing the spatial–spectral correlations within the generated background image. The final output of the background generation block is defined as $\mathbf{F}_{\text{bg}}^N\in\mathbb{R}^{H\times W\times c}$, where N is the number of background generation blocks and c is the number of channels.

E. Background Smoothing Block

Repeated usage of corner-visible convolutions may result in the grid effect, leading to the loss of continuity in the background information. To address the issue, conventional convolutions are exploited to smooth the background. The initial input of background smoothing block $\mathbf{F}_{\text{bs}}^N\in\mathbb{R}^{H\times W\times c}$ is

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obtained by skip connection

$$\mathbf{F}_{\mathrm{bs}}^{0} = \mathrm{concat}\left(\mathbf{F}_{\mathrm{bg}}^{N}, \mathrm{conv}_{1\times1}\left(\mathbf{F}_{\mathrm{bg}}^{N-1}\right)\right). \tag{7}$$

The skip connection serves to preserve detailed information from the network's initial stages, thereby enhancing the model's ability to simultaneously capture fine-grained details and high-level context. The background smoothing block consists of 3×3 convolution (with BN and ReLU) and 1×1 convolution (with BN and ReLU). The input of the ith background smoothing block can be represented as

$$\mathbf{F}_{\mathrm{bs}}^{i} = \mathrm{concat}\left(\mathrm{conv}_{1\times1}\left(\mathrm{conv}_{3\times3}\left(\mathbf{F}_{\mathrm{bs}}^{i-1}\right)\right)\mathrm{conv}_{1\times1}\left(\mathbf{F}_{\mathrm{bg}}^{N-1-i}\right)\right) \tag{8}$$

where $0 < i < N$. As a result, the final smoothed background $\mathbf{F}_{\mathrm{bs}}$ can be found as

$$\mathbf{F}_{\mathrm{bs}} = \mathrm{conv}_{1\times1}\left(\mathrm{conv}_{3\times3}\left(\mathbf{F}_{\mathrm{bs}}^{N-1}\right)\right). \tag{9}$$

Finally, the reconstructed background image is eventually obtained as $\hat{\mathbf{X}} = \mathrm{conv}_{1\times1}(\mathbf{F}_{\mathrm{bs}}) \in \mathbb{R}^{H\times W\times C}$, where the number of channels is the same with the original HSI. The final detected anomalies are identified as

$$\mathbf{D} = \left\|\mathbf{X} - \hat{\mathbf{X}}\right\|_2. \tag{10}$$

F. Loss Function

During the iterative training process, our objective is to ensure that the background pixels in the generated image closely resemble those in the original HSI image while simultaneously suppressing the reconstruction of abnormal pixels. Inspired by Auto-AD [23], the adaptive MSE loss is adopted for HAD. The specific calculation process is as follows:

$$e_{i,j} = \left\|\mathbf{x}_{i,j} - \hat{\mathbf{x}}_{i,j}\right\|_2 \tag{11}$$

$$\mathbf{E} = \left[e_{1,1}, \ldots, e_{1,W}; e_{2,1}, \ldots, e_{2,W}; \ldots; e_{H,1}, \ldots, e_{H,W}\right] \tag{12}$$

where $x_{i,j} \in \mathbb{R}^C$ denotes the (i, j) pixel vector in the original HSI image, $\hat{x}_{i,j} \in \mathbb{R}^C$ represents the corresponding reconstructed output vector, $e_{i,j}$ measures the difference between the original and reconstructed pixels, and $\mathbf{E}$ is the residual image composed of $e_{i,j}$. The weighted map W is then computed as follows:

$$w_{i,j} = \max(\boldsymbol{E}) - e_{i,j} \tag{13}$$

$$\boldsymbol{W} = \left[w_{1,1}, \ldots, w_{1,W}; w_{2,1}, \ldots, w_{2,W}; \ldots; w_{H,1}, \ldots, w_{H,W}\right]. \tag{14}$$

Consequently, the adaptive MSE loss is defined as

$$\mathcal{L}_m = \mathrm{MSE}\left(\mathbf{WX}, \mathbf{W}\hat{X}\right) = \sum_{i=1}^{H}\sum_{j=1}^{W}\left\|\left(\mathbf{x}_{i,j} - \hat{\mathbf{x}}_{i,j}\right)w_{i,j}\right\|_2. \tag{15}$$

The weighted map assigns lower weights to regions with significant reconstruction errors, effectively suppressing the contribution of anomalous pixels. To evaluate the quality of the reconstruction, we employ the adaptive MSE, which calculates the weighted squared difference between pixel intensities in the original and reconstructed HSIs. However, it is important to note that the adaptive MSE primarily focuses on pixel-level differences and may be overly sensitive to minor changes in pixel values, thereby overlooking perceptual differences and structural information present in the HSI data.

To address the limitations of the adaptive MSE loss in capturing structural information, we propose the adaptive SSIM loss, which incorporates spatial correlations to guide the reconstructed HSI with structural information. The adaptive SSIM loss evaluates the structural similarity between the weighted original and reconstructed HSIs by comparing local neighborhoods and measuring similarity based on perceived changes within these neighborhoods. Similar to the adaptive MSE loss, the adaptive SSIM loss aims to preserve the structural characteristics of the background while suppressing the reconstruction of anomalies during the training process. The adaptive SSIM loss is defined as follows:

$$\mathcal{L}_s = 1 - \mathrm{SSIM}\left(\mathbf{WX}, \mathbf{W}\hat{X}\right) \tag{16}$$

where SSIM is the structural similarity index of two images.

The final loss function is defined to consider both pixel-level differences and perceptual quality

$$\mathcal{L} = \mathcal{L}_m + \lambda\mathcal{L}_s \tag{17}$$

where the parameter λ is used to control the balance between the two loss functions.

Algorithm 1 CVNet

Input: hyperspectral image $\mathbf{X} \in \mathbb{R}^{\mathbf{H}\times\mathbf{W}\times\mathbf{C}}$, parameters:
1) receptive field size k and sub-kernel size k';
2) loss function term λ

Output: detection result $\mathbf{D} \in \mathbb{R}^{H\times W}$

```
Initiate the input noise X_o ∈ R^{H×W×C} and weight map W ∈ R^{H×W};
while iterations < 1000 do
    Network forward;
    if iterations %200 = 0 then
        Update the weight map using Eqs.11-14;
    end
    Calculate loss using Eqs.15-17;
    Network backward;
end
Obtain the reconstructed background image X̂ ∈ R^{H×W×C};
Calculate the detection result using Eq.10.
```

IV. Experimental Results and Analysis

The performance of CVNet was evaluated on multiple datasets to validate the effectiveness of the proposed method. Additionally, a comparison was made between our method and state-of-the-art HAD techniques, including RX [10], CRD [11], KIFD [44], GTVLRR [45], PCA-TLRSR [46], Auto-AD

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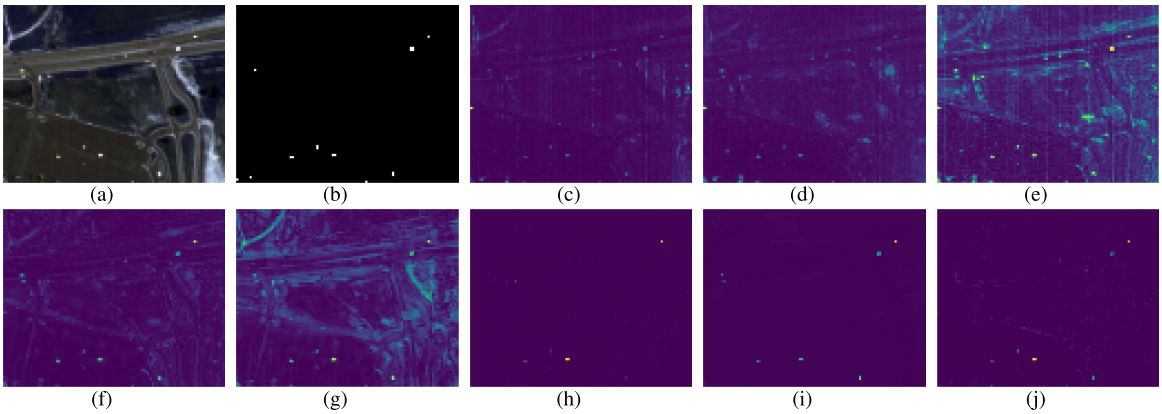


Fig. 4. HYDICE dataset and the anomaly detection results. (a) RGB pseudo-color image. (b) Ground truth. (c) RX. (d) CRD. (e) KIFD. (f) GTVLRR. (g) PCA-TLRSR. (h) Auto-AD. (i) LREN. (j) CVNet.

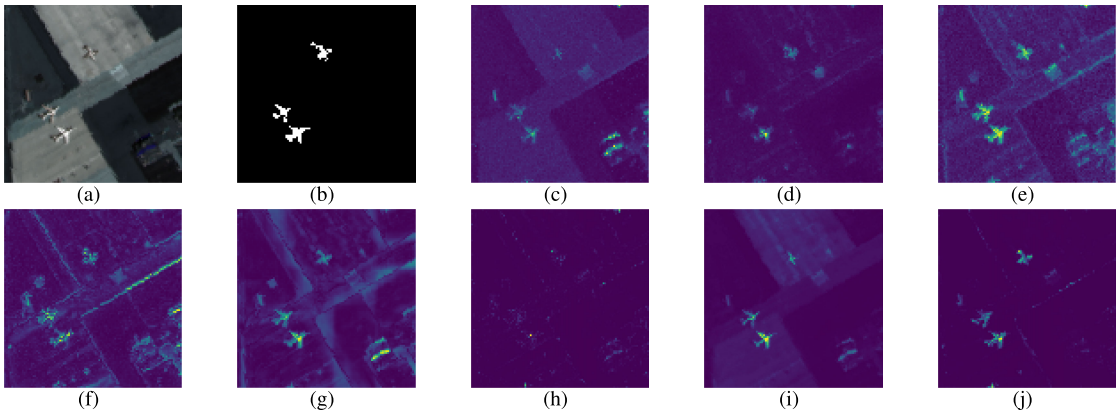


Fig. 5. San Diego I and the anomaly detection results. (a) RGB pseudo-color image. (b) Ground truth. (c) RX. (d) CRD. (e) KIFD. (f) GTVLRR. (g) PCA-TLRSR. (h) Auto-AD. (i) LREN. (j) CVNet.

[23], and LREN [47]. Furthermore, ablation experiments were conducted to assess the impact of the introduced corner-visible convolution and the incorporation of CBAM and SSIM loss functions.

A. Data Description

1) HYDICE Dataset: The HYDICE dataset was acquired using the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor [48] in a suburban residential area located in Michigan, USA. This dataset encompasses a spectral range of 400–2500 nm and has a spatial resolution of 3 m per pixel, along with a spectral resolution of 10 nm. The image size measures 80 × 100 pixels. Following the removal of bands with low signal-to-noise ratios (SNRs) (bands 1–4, 76, 87, 101–111, 136–153, and 198–210), a total of 162 spectral bands were retained for further analysis. Within the dataset, 17 pixels that represent synthetic vehicles serve as anomalies. These vehicle anomalies are spatially distributed within a range of 1 × 1–2 × 2 pixels for each vehicle. The background land cover types encompass vegetation, soil, water, and roads. The false-color image and the corresponding ground-truth image are depicted in Fig. 4(a) and (b).

2) San Diego Airport Dataset: The San Diego airport dataset was captured using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) at San Diego Airport, San Diego, CA, USA [49]. The HSI possesses a dimension of 400 × 400 pixels, featuring a spectral resolution of 10 nm and a total of 224 spectral bands spanning from 0.37 to 2.51 μm. After eliminating bands with low SNRs and water absorption bands (bands 1–6, 33–35, 97, 107–113, 153–166, and 221–224), a total of 189 bands were retained for experimentation purposes. Two sub-regions measuring 100 × 100 pixels were extracted from the HSI and utilized in the experiments, denoted as San Diego I and San Diego II. In each sub-image, the aircraft is considered the anomaly target, with the anomalies distributed spatially within a range of 6 × 6–11 × 14 pixels. The pseudo-color images and corresponding ground-truth images of the two sub-images are displayed in Figs. 5(a) and (b) and 6(a) and (b).

3) Pavia Dataset: The Pavia dataset was obtained using the Reflective Optics System Imaging Spectrometer (ROSSI-03) sensor in Pavia, Italy [13]. This HSI comprises 102 bands and possesses a spatial resolution of 1.3 m per pixel. For the experiment, a sub-image measuring 150 × 150 pixels was extracted from the HSI, and all bands were retained for subsequent analysis. Within this HSI, cars are regarded as anomaly targets, with their spatial distribution confined within a range of 4 × 4–5 × 5 pixels. The pseudo-color image and the corresponding ground-truth image of this HSI are depicted in Fig. 7(a) and (b).

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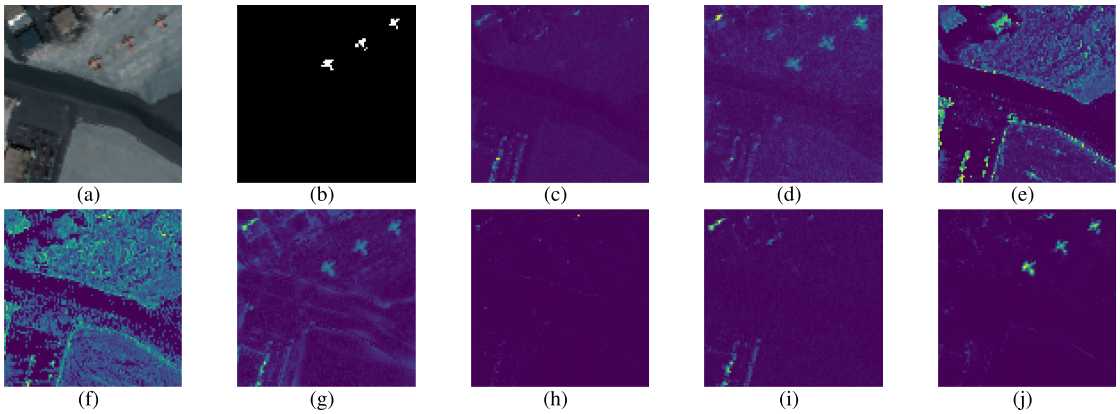


Fig. 6. San Diego II and the anomaly detection results. (a) RGB pseudo-color image. (b) Ground truth. (c) RX. (d) CRD. (e) KIFD. (f) GTVLRR. (g) PCA-TLRSR. (h) Auto-AD. (i) LREN. (j) CVNet.

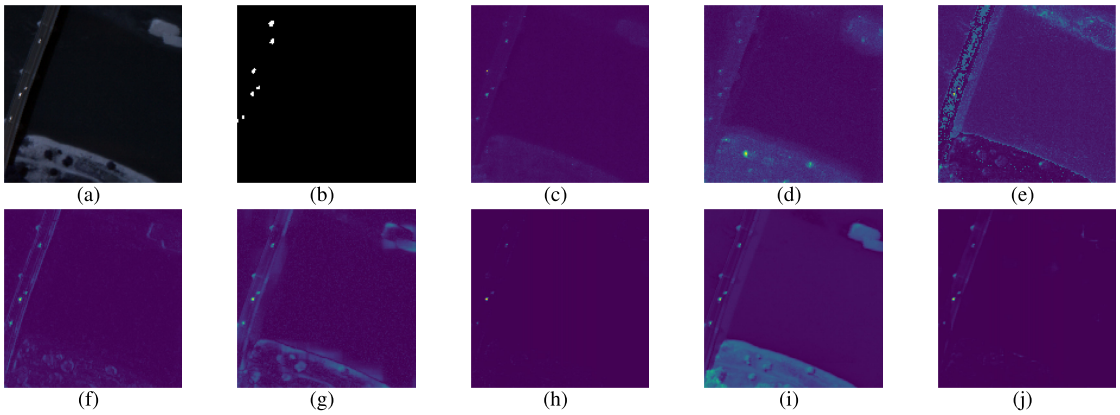


Fig. 7. Pavia dataset and the anomaly detection results. (a) RGB pseudo-color image. (b) Ground truth. (c) RX. (d) CRD. (e) KIFD. (f) GTVLRR. (g) PCA-TLRSR. (h) Auto-AD. (i) LREN. (j) CVNet.

B. Parameter Settings

In the experiments, the CVNet model incorporates five repeated background generation and background smoothing modules. The training of the network is optimized using the Adam optimizer, with a learning rate of 0.1. The spectral band reduction ratio is set to 0.6. The iterative learning process is performed for 1000 epochs, and the weights in the loss function are adaptively initialized every 200 epochs. The evaluation of our method employs the area under the curve (AUC) of the receiver operating characteristic (ROC) curve, which considers both the performance detection (P_d) and performance false (P_f) rates. A higher $\mathrm{AUC}_{(P_d,P_f)}$ indicates a stronger capability of the detector to identify abnormal samples, while a lower $\mathrm{AUC}_{(P_f,\tau)}$ suggests fewer false alarms generated by the detector.

C. Analysis of the Detection Results of HYDICE

The detection maps of the eight HAD methods on the HYDICE dataset are shown in Fig. 4. In the HYDICE dataset, characterized by small-sized anomalies, each anomaly typically consists of 1×1 to 2×2 pixels. Furthermore, the distribution of these abnormal objects is relatively scattered. As shown in Fig. 4(j), the CVNet demonstrates strong performance in background suppression, effectively removing unwanted anomaly information from the input data. Furthermore, the detection results obtained by CVNet closely align with the ground truth, indicating accurate and reliable distinguishing anomalies from the background.

Traditional anomaly detection methods based on statistical learning (e.g., RX, CRD) are often susceptible to background interference, resulting in incomplete separation between anomalies and the background. This limitation leads to significant false positive detections, where normal background elements are mistakenly classified as anomalies. One of the reasons is that the statistical learning methods lie in their reliance on assumptions about the underlying data distribution, which may not hold in complex real-world scenarios. The prior assumptions can restrict their ability to effectively distinguish anomalies from the background and contribute to an increased false positive rate.

By capturing complex relationships and dependencies, deep learning-based HAD methods exhibit a greater capacity to handle background reconstruction, resulting in improved separation between anomalies. For example, the Auto-AD method exhibits robust resistance to background interference and effectively separates anomalies from the background. However, it still experiences occasional instances of missed detections, where certain anomalies may go undetected. Similarly, the LREN method, while constructing dictionaries,

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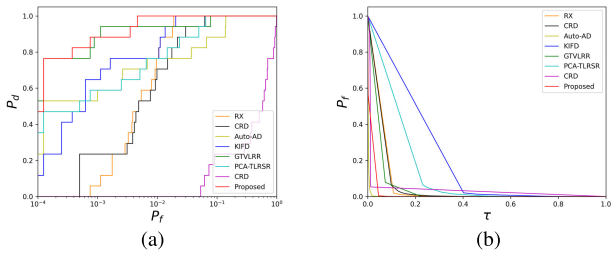


Fig. 8. ROC curves (a) (P_d, P_f) and (b) (P_f, τ) on the HYDICE dataset.

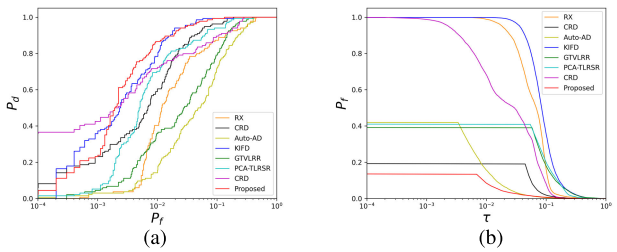


Fig. 9. ROC curves (a)(P_d, P_f) and (b) (P_f, τ) on San Diego I.

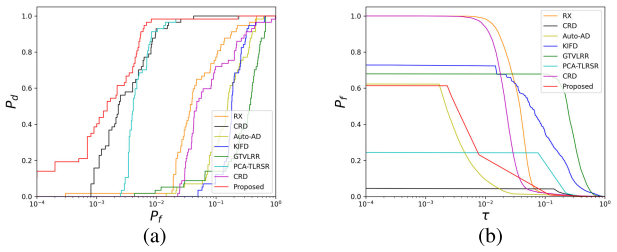


Fig. 10. ROC curves (a) (P_d, P_f) and (b) (P_f, τ) on San Diego II.

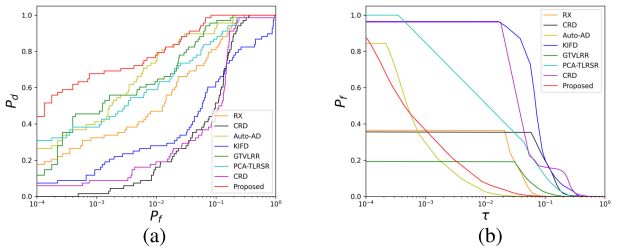


Fig. 11. ROC curves (a) (P_d, P_f) and (b) (P_f, τ) on the Pavia dataset.

can sometimes erroneously confuse certain anomalies with the background, resulting in a decreased detection rate. From the detection maps, it is evident that the CVNet surpasses the performance of Auto-AD and LREN on the HYDICE dataset, underscoring its effectiveness in background suppression and accurate anomaly detection. The notable improvement can be attributed to CVNet's avoidance of downsampling and upsampling operations during the feature extraction process, which mitigates information loss arising from variations in feature scales. By carefully selecting suitable parameters for corner-visible convolutions, CVNet effectively suppresses the reconstruction of anomalies during the background reconstruction phase, thereby achieving a more accurate approximation of the ground truth for the separated anomalies.

From the ROC curves depicted in Fig. 8(a) and (b), it is evident that CVNet achieves the highest $\mathrm{AUC}_{(P_d,P_f)}$ score, indicating its superior performance in effectively separating the background from anomalies. Additionally, CVNet demonstrates the ability to achieve a low $\mathrm{AUC}_{(P_f,\tau)}$ level, indicating a low false alarm rate. These results highlight the effectiveness of our proposed method in achieving a high detection rate while minimizing false alarms.

D. Analysis of the Detection Results of San Diego Airport Datasets

As shown in Figs. 5(b) and 6(b). Both San Diego I and II exhibit three identified aircraft anomalies. In San Diego I, the anomalies have sizes ranging from 10×11 to 11×14 pixels, while in San Diego II, the anomalies range in size from 6×6 to 7×7 pixels. The anomalies in both datasets generally possess larger sizes and a more concentrated distribution. The presence of larger-sized and concentrated anomalies poses higher demands on anomaly detection methods, especially the CNN-based deep learning HAD methods.

In San Diego I, due to the significant spectral differences between anomalies and the background, all methods except for Auto-AD can locate anomalous targets. The reason for the ineffectiveness of Auto-AD lies in its excessively powerful feature extraction capability with CNN to capture high-level semantic features of the anomalies, resulting in a low-quality separation process. Due to the presence of large-scale concentrated anomalies, the convolutional operation, when reconstructing these pixels, encompasses a considerable amount of adjacent abnormal information within its local receptive field. This results in a failure in the background reconstruction process. In contrast, the proposed corner-visible convolution masks the neighboring pixels of the central pixel to be reconstructed within the local receptive field. This helps to minimize the impact of probability correlation between abnormal pixels, enabling the reconstruction of abnormal pixels in the original image as background.

When compared to other methods, our proposed method demonstrates a significantly lower false alarm rate. This can be observed from the ROC curves presented in Fig. 9(a) and (b), where CVNet exhibits the lowest false alarm rate at a detection rate of 100%. Moreover, CVNet achieves the highest $\mathrm{AUC}_{(P_d,P_f)}$ value, indicating its superior ability to maintain a high detection rate while minimizing false alarms. Additionally, CVNet also achieves the lowest $\mathrm{AUC}_{(P_f,\tau)}$ value, indicating its effectiveness in achieving a low false alarm rate. The improved performance, as indicated by the high $\mathrm{AUC}_{(P_d,P_f)}$ value and the low $\mathrm{AUC}_{(P_f,\tau)}$ value, showcases the effectiveness and reliability of CVNet in accurately identifying anomalies while maintaining a low rate of false alarms.

In San Diego II, anomalies exhibit spectral features that are similar to buildings. This similarity poses a challenge for traditional methods such as RX, Auto-AD, and LREN in effectively separating anomalies from the background. Additionally, the detection results of KIFD and GTVLRR methods are also adversely affected by background interference, leading to a significant number of false alarms. Although the CRD and PCA-TLRSR methods can locate the anomalies more clearly,

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TABLE I
$AUC_{(P_d,P_f)}$ ↑ OF DIFFERENT METHODS

Methods	RX	CRD	KIFD	GTVLRR	PCA-TLRSR	Auto-AD	LREN	CVNet
Hydice	0.9932	0.9885	0.9964	0.9952	0.9900	0.9971	0.8187	**0.9994**
SanDiego I	0.9513	0.9827	0.9922	0.9429	0.9784	0.9141	0.9696	**0.9930**
SanDiego II	0.9219	**0.9950**	0.7874	0.6464	0.9905	0.8031	0.8686	0.9935
Pavia	0.9538	0.8970	0.8162	0.9803	0.9646	0.9813	0.8897	**0.9907**
Average	0.9551	0.9658	0.8981	0.8912	0.9809	0.9239	0.8867	**0.9942**

The best and second best results are highlighted and underlined.

TABLE II
$AUC_{(P_f,\tau)}$ ↓ OF DIFFERENT METHODS

Methods	RX	CRD	KIFD	GTVLRR	PCA-TLRSR	Auto-AD	LREN	CVNet
Hydice	0.0567	0.0561	0.2083	0.0457	0.1285	**0.0027**	0.0316	0.0137
SanDiego I	0.0704	0.0312	0.1005	0.0689	0.0672	0.0071	0.0402	**0.0069**
SanDiego II	0.0399	0.0769	0.1117	0.2359	0.0690	**0.0112**	0.0282	0.0191
Pavia	0.0199	0.0609	0.0851	0.0246	0.0457	**0.0016**	0.0727	0.0032
Average	0.0467	0.0563	0.1264	0.0938	0.0776	**0.0057**	0.0432	0.0107

The best and second best results are highlighted and underlined.

they still encounter issues with false detections. However, CVNet successfully addresses these challenges by leveraging the CBAM. Through the use of channel-wise attention, CBAM considers global contextual information and learns to focus on important spectral features while attenuating less informative ones. This mechanism enables the network to emphasize discriminative spectral characteristics, enhancing its ability to differentiate objects with similar spectral curves.

As depicted in the ROC curves presented in Fig. 10(a) and (b), the KIFD method achieves the highest $AUC_{(P_d,P_f)}$ score, indicating its superior overall performance in terms of detection rate and false positive rate. However, it is noteworthy that CVNet consistently demonstrates better anomaly separation performance compared to KIFD. Although KIFD achieves a high $AUC_{(P_d,P_f)}$ score, which reflects its ability to maintain a good balance between detection rate and false positive rate, CVNet excels in terms of accurately separating anomalies from the background, which highlights the strength of CVNet in achieving precise and reliable anomaly detection outcomes.

E. Analysis of the Detection Results of Pavia Dataset

As depicted in Fig. 7, the anomalies present in the Pavia dataset are characterized by sizes ranging from 4×4 to 5×5 pixels. These anomalies exhibit a concentrated distribution pattern, appearing in specific localized regions. Furthermore, the Pavia dataset comprises a limited number of distinct land cover types, resulting in a relatively simple background.

In this particular scenario, all methods are capable of locating the anomalies in the Pavia dataset. However, traditional statistical methods still suffer from significant background interference, which hampers their ability to accurately detect anomalies. The Auto-AD, while successful in reducing false alarms, has a strong reconstruction capability that hinders the effective separation of anomaly targets. On the other hand, LREN performs reasonably well in separating anomalies but mistakenly identifies nearshore objects as anomalies as well. To address the issue, CVNet introduces the SSIM to enhance the effectiveness of HAD. The integration of adaptive SSIM loss ensures the preservation of intricate structural details during the detection process. This preservation of fine-grained information plays a crucial role in achieving a more precise and reliable background reconstruction process, ultimately leading to enhanced anomaly detection performance.

As depicted in the ROC curves in Fig. 11(a) and (b), CVNet achieves the highest $AUC_{(P_d,P_f)}$ level and a lower $AUC_{(P_f,\tau)}$ value compared to other methods. This indicates the effectiveness of our method in detecting anomalies in the Pavia dataset. The high $AUC_{(P_d,P_f)}$ level reflects the capability of CVNet in achieving a good balance between the detection rate and false positive rate, while the lower $AUC_{(P_f,\tau)}$ value suggests a reduced false alarm rate.

F. Overall Comparisons

Overall, CVNet has exhibited exceptional performance across a diverse range of anomaly detection datasets, encompassing both point anomaly and structured anomaly scenarios. Tables I and II provide a comprehensive comparison of the $AUC_{(P_d,P_f)}$ values and $AUC_{(P_f,\tau)}$ values for eight different HAD methods, including CVNet. The results unequivocally demonstrate the superior performance of CVNet, as evidenced by its higher $AUC_{(P_d,P_f)}$ values and average values in HYDICE, San Diego, and Pavia datasets. Although CVNet slightly lags behind the CRD method in terms of $AUC_{(P_d,P_f)}$ values in San Diego II, it is important to note that the CRD method exhibits significantly higher $AUC_{(P_f,\tau)}$ values compared to CVNet.

While CVNet may not achieve the highest $AUC_{(P_d,P_f)}$ values in San Diego II compared to the CRD method, it is crucial to consider the corresponding $AUC_{(P_f,\tau)}$ values. These values reflect the false alarm rate at a fixed detection threshold and are equally important in evaluating the performance of anomaly detection methods. In this aspect, CVNet performs better than the CRD method, further emphasizing its ability to effectively suppress false alarms and maintain a low false positive rate. Furthermore, CVNet consistently demonstrates

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TABLE III
$AUC_{(P_d,P_f)}$ ↑ VALUES OF THREE DIFFERENT CONVOLUTIONS ON THREE DATASETS

	Convolution	Blind-Spot	Corver-Visible
Hydice	0.9860	0.9314	**0.9994**
SanDiego I	0.9251	0.9390	**0.9930**
SanDiego II	0.7272	0.9384	**0.9935**
Pavia	0.9783	0.9637	**0.9907**
Average	0.9042	0.9431	**0.9942**

The best results are highlighted.

TABLE IV
$AUC_{(P_d,P_f)}$ ↑ VALUES FOR DIFFERENT k AND k' VALUES TAKEN ON THE HYDICE DATASET

k' \ k	3	5	7	9	11
1	0.9913	0.9863	**0.9994**	0.9926	0.9601
2	–	0.9982	0.9762	0.9687	0.9752
3	–	–	0.9740	0.9867	0.9832

The best results are highlighted.

TABLE V
$AUC_{(P_d,P_f)}$ ↑ VALUES FOR DIFFERENT k AND k' VALUES TAKEN ON SAN DIEGO I

k' \ k	3	5	7	9	11
1	0.9407	0.9741	0.9803	0.9896	0.9747
2	–	0.9858	**0.9930**	0.9733	0.9450
3	–	–	0.9887	0.9616	0.9549

The best results are highlighted.

TABLE VI
$AUC_{(P_d,P_f)}$ ↑ VALUES FOR DIFFERENT k AND k' VALUES TAKEN ON SAN DIEGO II

k' \ k	3	5	7	9	11
1	0.9750	0.8928	0.9541	0.9566	0.9068
2	–	**0.9935**	0.9453	0.9305	0.7971
3	–	–	0.9442	0.6111	0.7708

The best results are highlighted.

lower $AUC_{(P_f,\tau)}$ values across the HYDICE, San Diego, and Pavia datasets. On the other hand, the relatively lower performance of the Auto-AD method can be attributed to its inability to effectively separate anomalies from the background in structured anomaly datasets such as San Diego I, San Diego II, and Pavia. The detection results of Auto-AD neither accurately identify anomalies nor capture the background information, indicating the need to simultaneously achieve high $AUC_{(P_d,P_f)}$ levels and low $AUC_{(P_f,\tau)}$ levels for effective detection outcomes. The consistent performance of CVNet in obtaining lower $AUC_{(P_f,\tau)}$ values and higher $AUC_{(P_d,P_f)}$ values reflects its ability to maintain a low false positive rate, thus enhancing the reliability of anomaly detection outcomes.

In summary, the CVNet stands out from other HAD methods by its ability to adapt to all scenes, further validating its effectiveness in HAD tasks. Unlike other HAD methods that are designed for specific single scenes or limited anomaly types, CVNet offers a more comprehensive and flexible approach to anomaly detection. Its capability to detect and differentiate between different anomaly types, ranging from isolated point anomalies to more complex structured anomalies, makes it a robust and versatile tool in the field of HAD. The exceptional background reconstruction capabilities of CVNet can be attributed to the integration of multiple key components. These include a network structure that avoids upsampling or downsampling operations, the utilization of corner-visible convolution to eliminate the correlation of anomalous pixels, the spectral processing approach based on CBAM, and the incorporation of the adaptive SSIM loss function.

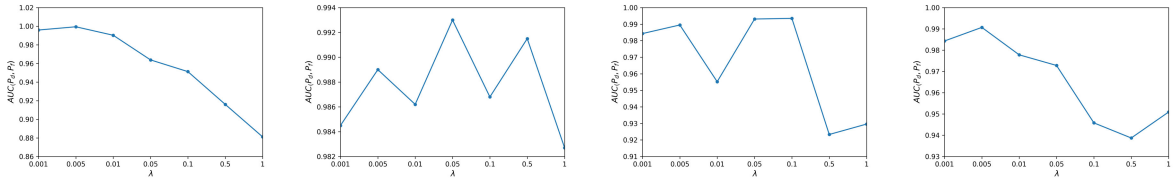
G. Effectiveness of Corner-Visible Convolution

In this part, experiments were conducted to verify the effectiveness of corner-visible convolution. In HAD tasks, different convolutional operations are commonly used to aggregate neighborhood information for background reconstruction. Table III presents the $AUC_{(P_d,P_f)}$ values on HYDICE, San Diego, and Pavia datasets by performing different convolutions, including conventional convolution, blind-spot convolution, and our proposed corner-visible convolution. From Table III, it can be observed that the $AUC_{(P_d,P_f)}$ values of the corner-visible convolution are significantly higher than those of the conventional and blind-spot convolutions on all the datasets, which demonstrates the superiority of the proposed corner-visible convolution in HAD. Fundamentally, the difference among these three types of convolutions lies in their utilization of local spatial information within the receptive field.

Conventional convolution operates by capturing the correlations among all pixels within its receptive field to extract high-level semantic features, originally intended for classification tasks. However, when a central pixel is an anomaly, it is highly unlikely to be reconstructed as background due to identity learning, leading to failures in HAD tasks. Blind-spot convolution reduces the influence of the central pixel during the background reconstruction process by erasing the information from the central pixel to forestall network learning identity. However, blind-spot convolution is specifically designed for denoising tasks and performs well for scattered anomalies but is not suitable for large anomaly scenes. When the majority of pixels in the neighborhood of a central pixel are anomalies, this central pixel is unlikely to be reconstructed as background, leading to failures in HAD tasks. Therefore, the performance of blind-spot convolution also deteriorates when dealing with larger anomaly sizes, as evidenced by the results on the San Diego dataset presented in Table III.

The corner-visible convolution, when encountering anomalous pixels, mitigates the correlation of the abnormal pixels by masking the nearest neighboring pixels (which are likely to be anomalies as well). Then the correlation of background pixels is exploited for background reconstruction by utilizing the corner pixels in the far end of the local receptive field (which are likely to be the background). As a result, the corner-visible convolution is capable of reconstructing a relatively pure background image, enabling the detection results to achieve high $AUC_{(P_d,P_f)}$.

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(a) (b) (c) (d)

Fig. 12. $\mathrm{AUC}_{(P_d,P_f)}$ values for different λ taken on three datasets. (a) HYDICE. (b) San Diego I. (c) San Diego II. (d) Pavia.

TABLE VII

$\mathrm{AUC}_{(P_d,P_f)}$ ↑ VALUES FOR DIFFERENT k AND k' VALUES TAKEN ON THE PAVIA DATASET

k' \ k	3	5	7	9	11
1	0.9587	0.9502	**0.9907**	0.9687	0.9678
2	–	0.9651	0.9665	0.9704	0.9753
3	–	–	0.9666	0.9761	0.9823

The best results are highlighted.

TABLE VIII

AVERAGE $\mathrm{AUC}_{(P_d,P_f)}$ ↑ AND $\mathrm{AUC}_{(P_f,\tau)}$ ↓ VALUES OF ABLATION EXPERIMENTS ON THREE DATASETS

	CBAM	SSIM	$AUC_{(P_d,P_f)}$ ↑	$AUC_{(P_f,\tau)}$ ↓
CVNet	✓	✓	**0.9942**	**0.0107**
CVNet-C		✓	0.9840	0.0133
CVNet-S	✓		0.9807	0.0114
CVNet-CS			0.9740	0.0166

The best results are highlighted.

H. Influence of the Parameters of Corner-Visible Convolution

In this part, we investigate the influence of parameter settings for corner-visible convolution, including the size of receptive field k and the size of sub-kernel k'. Tables IV–VII display the $\mathrm{AUC}_{(P_d,P_f)}$ values of CVNet for detection results on the HYDICE, San Diego, and Pavia datasets, with varying sizes of receptive fields (k) and sub-kernels (k'). It is evident that the optimal choices of k and k' vary for different anomaly types. In the case of larger anomalies observed in San Diego I and San Diego II, selecting larger sub-kernels significantly enhances the $\mathrm{AUC}_{(P_d,P_f)}$ values. Conversely, for smaller anomalies in HYDICE and Pavia, employing smaller sub-kernels yields higher accuracy. Moreover, when there is a significant disparity between anomalies and background features, opting for a broader range of receptive fields effectively suppresses anomaly reconstruction. However, excessively large receptive fields weaken the spatial correlation between sampled information from different sub-kernels, which hinders background reconstruction.

While CVNet demonstrates strong performance in detecting anomalies at various scales, it is important to highlight that the configuration of hyperparameters for corner-visible convolution receptive fields is necessary before each detection. Thus, when dealing with situations where multiple scale anomalies coexist, the detection performance may be affected. As a potential solution, a multiscale feature fusion approach may offer assistance for anomaly detection in this scenario [50].

I. Influence of the Balance Parameter λ

The parameter λ is used to balance the adaptive MSE loss and adaptive SSIM loss, thereby controlling the influence of pixel differences and structural differences on the detection results. The parameter λ is selected from $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1\}$. Fig. 12(a)–(d) demonstrate the impact of λ on the detection results of HYDICE, San Diego I, San Diego II, and Pavia, respectively. As shown in Fig. 12(a), when λ is small, the detection results exhibit higher $\mathrm{AUC}_{(P_d,P_f)}$ values, reaching the highest at $\lambda = 0.005$. This indicates that the anomalies in HYDICE do not possess evident structural characteristics, and the anomaly detection process mainly relies on the pixel differences between the background image and the original image. As depicted in Fig. 12(b) and (c), the $\mathrm{AUC}_{(P_d,P_f)}$ values of structural anomalies are highest when λ is within the range of 0.05–0.01, showing that the structural loss SSIM plays a role in the detection process of structural anomalies. However, when $\lambda > 0.1$, excessively high weights of structural differences may lead to confusion between background objects resembling anomalies and actual anomaly targets, thus affecting the detection accuracy. As shown in Fig. 12(d), the $\mathrm{AUC}_{(P_d,P_f)}$ level reaches its peak when $\lambda = 0.005$. This is because the Pavia image has a larger scale, and the proportion of anomalous samples is relatively small. Therefore, it is necessary to choose a smaller value of λ on Pavia to achieve optimal performance.

J. Ablation Study

In this part, experiments were conducted to verify the effectiveness of CBAM and SSIM loss. Table VIII provides the average $\mathrm{AUC}_{(P_d,P_f)}$ values and average $\mathrm{AUC}_{(P_f,\tau)}$ values for four abnormal images, with and without CBAM or SSIM. The results presented in Table VIII highlight the significant improvements in anomaly detection accuracy achieved by introducing CBAM and SSIM. Furthermore, CBAM demonstrated its capability to enhance the $\mathrm{AUC}_{(P_d,P_f)}$ values while maintaining low $\mathrm{AUC}_{(P_f,\tau)}$ values, by incorporating global context information and selectively focusing on important spectral features to enhance the spatial–spectral features, which is advantageous for HAD. Additionally, the adaptive SSIM loss function preserved intricate structural details during the detection process, ensuring precise and reliable background reconstruction.

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V. Conclusion

In conclusion, this article introduces an end-to-end CVNet for unsupervised HAD that demonstrates remarkable versatility and applicability across various anomaly types. The incorporation of corner-visible convolution allows to harness statistical dependencies within the receptive field while effectively eliminating spatial correlations with potential anomalies during background generation. Moreover, to address the grid effect, we employ a background smoothing module based on conventional convolution. Additionally, the use of adaptive MSE and SSIM losses effectively suppresses anomaly reconstruction errors, resulting in a robustly reconstructed background map.

In future work, we will incorporate large segmentation models, such as SAM [51], to guide HAD tasks.

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Leiquan Wang (Member, IEEE) received the Ph.D. degree in communication and electrical systems from the Beijing University of Posts and Telecommunications (BUPT), Beijing, China, in 2016.

Now, he is a Lecturer at the College of Computer and Communication Engineering, China University of Petroleum (East China), Qingdao, China. His research interests include multimodal fusion, image/video caption, and hyperspectral analysis.



Yunfeng Wang received the bachelor's degree from the China University of Petroleum, Qingdao, China, in 2021. He is currently pursuing the master's degree with the College of Computer Science and Technology, China University of Petroleum (East China), Qingdao.

His research interests include unsupervised hyperspectral anomaly detection.



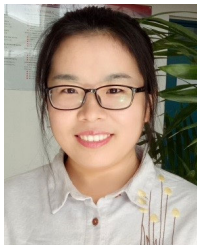
Zhongwei Li received the Ph.D. degree from the China University of Petroleum, Qingdao, China, in 2011.

He is currently a Professor with the College of Oceanography and Space Informatics, China University of Petroleum. His research interests include remote sensing image processing and ocean numerical forecasting and cloud computing.



Chunlei Wu (Member, IEEE) received the Ph.D. degree majoring in computer application technology from the Ocean University of China, Qingdao, China, in 2014.

He is a Male Associate Professor at the College of Petroleum (East China), Qingdao. Currently, he works at the University of Victoria, Victoria, BC, Canada, as a Visiting Scholar. He has authored and coauthored more than 30 journals and conference papers and textbooks. His research interests include image and video processing and machine learning.



Mingming Xu (Member, IEEE) received the B.S. degree in surveying and mapping engineering from the China University of Petroleum, Qingdao, China, in 2011, and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China, in 2016.

She is currently an Assistant Professor with the College of Oceanography and Space Informatics, China University of Petroleum. Her research interests include hyperspectral image processing and intelligent computation.



Mingwen Shao received the M.S. degree in mathematics from Guangxi University, Guangxi, China, in 2002, the Ph.D. degree in applied mathematics from Xi'an Jiaotong University, Xi'an, China, in 2005, and the Ph.D. degree in control science and engineering from Tsinghua University, Beijing, China, in 2008.

He is a Professor and a Doctoral Supervisor with the China University of Petroleum (East China), Qingdao, China. His research interests include rough sets, data mining, machine learning, and generative adversarial learning.

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